Sai Srikar Chowdary Pattipati

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Summary

Computer Science Master's (3.9 GPA, UMass Boston, May 2026) in Machine Learning + Natural Language Processing. Built AI evaluation platform (FastAPI, React, Anthropic Claude), 91.8% DistilBERT classifier, vital-stream Kafka pipeline (Spring Boot, 5k events/sec), 500+ LeetCode. Teaches LangChain, RAG, agents to 55 students.

Education

Master of Science in Computer Science — Aug 2024 - May 2026 (Expected)
University of Massachusetts Boston — GPA: 3.9/4.0

Bachelor of Engineering in Computer Science — Aug 2020 - May 2024
RMK Engineering College — GPA: 3.38/4.0

Experience

Teaching Fellow / Instructor - IT110 (Fall 2025) & CS188sl (Spring 2026) — Sep 2025 - May 2026
University of Massachusetts Boston — *Boston, MA*

- Designed and delivered IT110 (fall) and CS188sl (spring) to 55 students over 2 semesters, covering Python, Java OOP, REST APIs, algorithms, CS theory, and AI tooling.
- Introduced 25 students to LLMs (LangChain with RAG and agents, n8n workflow automation) through hands-on demos, bridging classroom instruction with current LLM-engineering practices.
- Covered software-engineering practice for 25 spring students across 2 sections: Java OOP, REST APIs, code reviews via Git/GitHub PRs, and SQL-injection awareness.

Projects

AI-Powered Job Search & Resume Tailoring Platform [GitHub]
FastAPI, React, SQLAlchemy, Anthropic Claude API, LaTeX, Jinja2, Tailwind CSS
- Built a full-stack AI product (FastAPI, React) aggregating listings from 5+ job boards, auto-extracting JDs from URLs via LLM and scoring resume-match 1-10.
- Designed an LLM evaluation pipeline emitting structured JSON metadata; deterministic Jinja2 to LaTeX to PDF rendering eliminates a class of LLM compile failures via automated guardrails.
- Built a chat-based resume refinement UX with conversation history, per-message snapshots, and one-click revert; enables human-in-the-loop iteration on model outputs.

vital-stream — Spring Boot + Kafka Biometric Event Pipeline [GitHub]
Java 21, Spring Boot 3, Spring Kafka, Spring Data JPA, Apache Kafka (KRaft, RF=3), PostgreSQL 16, Docker Compose
- Built vital-stream, a 3-broker Kafka pipeline (Spring Boot, Postgres, Docker Compose) ingesting biometric events at 5k/sec with sub-millisecond producer latency and zero consumer lag under sustained.

Mental Health Text Classification (NLP)
Python, PyTorch, DistilBERT, Scikit-learn, Hugging Face Transformers, pandas, Statistics
- Built and fine-tuned a healthcare NLP model for mental-health text classification (ADHD, depression, OCD, PTSD) from Reddit posts, achieving 91.8% accuracy with DistilBERT on a held-out test set.
- Conducted model evaluation by benchmarking Naive Bayes, feedforward neural networks, and DistilBERT; transformers outperformed traditional approaches by 15+ percentage points.

Cloud Resume Challenge [GitHub]
Terraform, AWS (S3, CloudFront, Lambda, DynamoDB, API Gateway, Route53), GitHub Actions, Python
- Provisioned full AWS stack (S3, CloudFront, Lambda, DynamoDB, IAM) via Terraform IaC with a GitHub Actions CI/CD pipeline for reproducible deployments.

Technical Skills

Machine Learning and AI: Machine Learning, Natural Language Processing, LLMs, RAG, Model Building, Model Evaluation, PyTorch, Hugging Face Transformers, Scikit-learn, pandas, Statistics, Anthropic Claude API
Languages: Python, Java, JavaScript, SQL, C/C++
Backend and APIs: FastAPI, REST APIs, Spring Boot, Spring Data JPA, Spring Kafka, Node.js, Express.js, Jinja2
Databases and Cloud: PostgreSQL, MySQL, SQLite, MongoDB, SQLAlchemy, Alembic, AWS (EC2, ECS, ALB, S3, RDS, Lambda, IAM)
Infrastructure and DevOps: Docker, Kubernetes, Terraform, Infrastructure as Code, GitHub Actions, Prometheus, Grafana, Linux, Git, Agile
CS Fundamentals: Data Structures, Algorithms, 500+ LeetCode, System Design, Object-Oriented Design, Unit Testing
Core Competencies: Ability to work independently, Collaboration, Teamwork, Communication, Accuracy, Thoroughness, Drive, Problem solving, Goal setting